Listening to the Pipes

Synergizing IoT, Machine Learning, and Geospatial Kriging for Proactive Leak Detection and Predictive Maintenance in Complex Urban Water Networks

Nishi Kant Mandal
nishim.ce22.063@bitsindri.ac.in
(*UG, Department of Civil Engineering, BIT Sindri*)

Anand Kumar Paswan
anandkp22@gmail.com
(*UG, Department of Mechanical Engineering, BIT Sindri*)

Asif Gulmozzam Fayazi
asifgulmozzamfayazi@gmail.com
(*UG, Department of Chemical Engineering, BIT Sindri*)

Problem

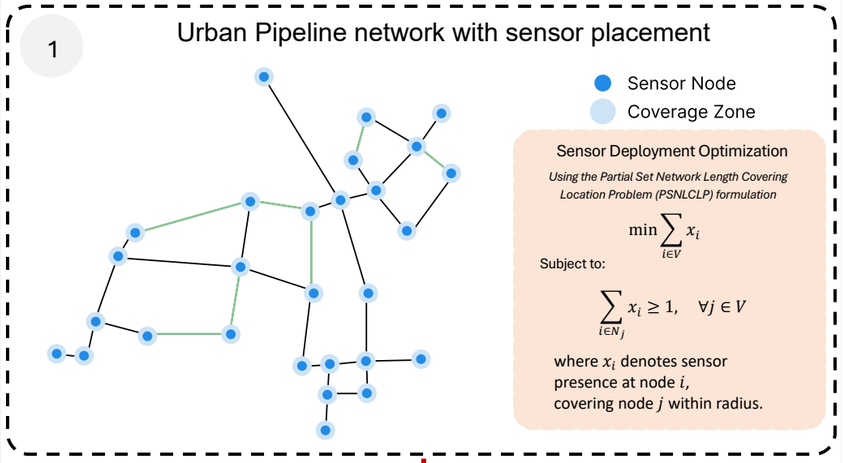
Urban water distribution networks face significant challenges, including extensive resource wastage due to leaks, which are further complicated by unpredictable demand fluctuations. These issues have broader implications for water sustainability and network efficiency. Despite existing research efforts, the dynamic nature of these networks remains a hurdle for effective leak detection, highlighting the need for innovative solutions that can adapt to ever-changing conditions.

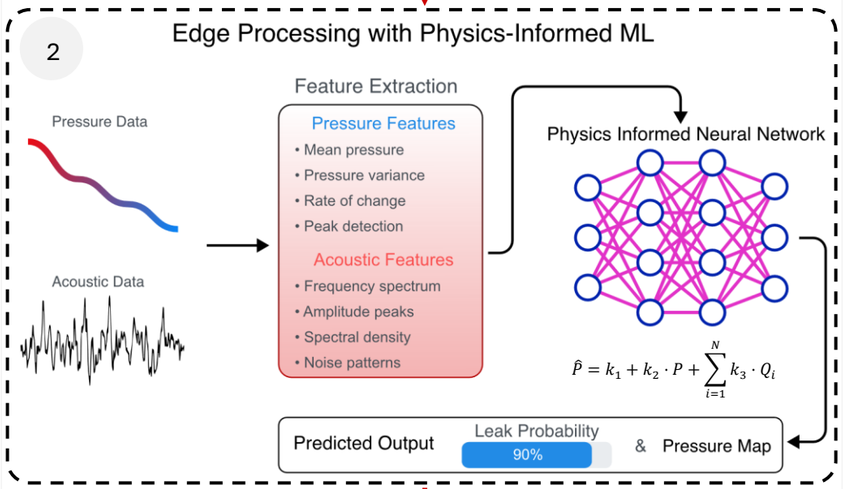
Background

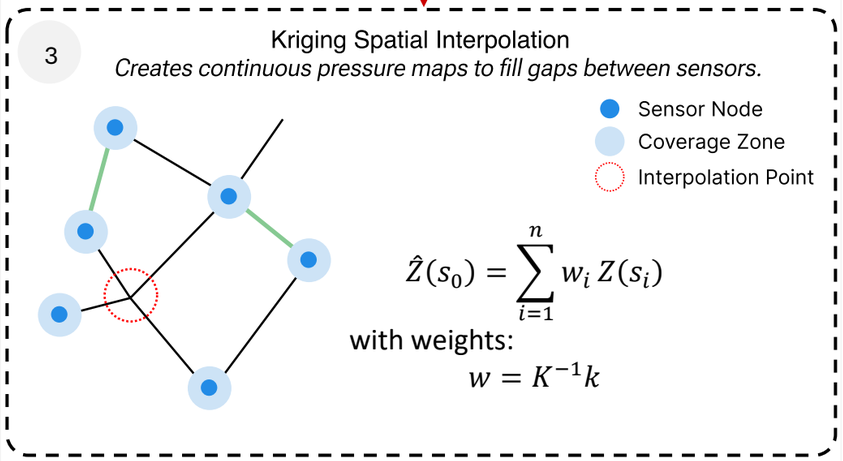
Traditional fixed leak detection systems often yield suboptimal results due to inflexibility in evolving network conditions, leading to undetected leaks or false alarms. This research leverages *Internet of Things* (IoT), *Machine Learning*, and *spatial interpolation* breakthroughs, building on existing studies [1]-[3], to introduce a Scalable, Real-Time Leak Detection Paradigm that responds to environmental and demand variability, enhancing Detection Precision and Efficiency.

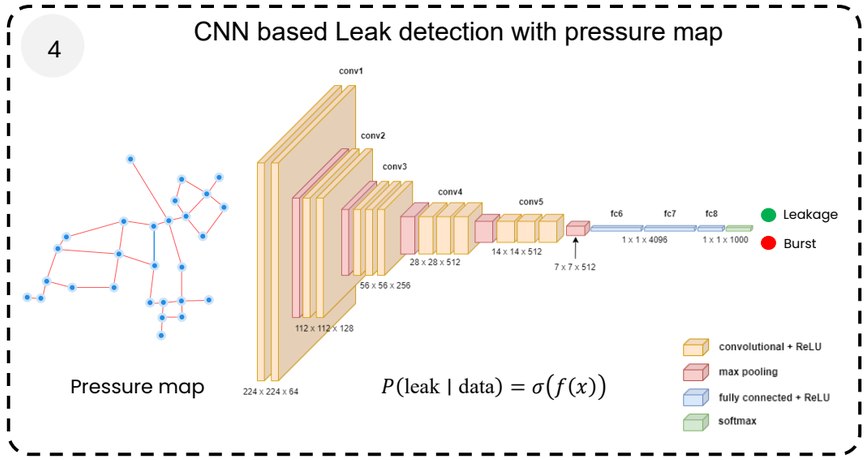
Approach

Our technical framework transforms water leak detection in dynamic urban hydraulic networks by integrating: Pressure & Acoustic Sensor Data (placed via Topological Optimization ①), Edge-based Physics-Informed Machine Learning ② for real-time anomaly detection Ordinary Kriging ③ for capturing spatial pressure autocorrelation. These converged insights power a CNN-based Probabilistic Predictive Maintenance ④ Scheduler, elevating the precision and speed of leak detection and response.









Implementation

Phase 1: Network Survey and Initial Deployment: Phase 1 involves mapping the network layout and deploying an initial sensor array for comprehensive coverage. Leak Detection Sensor Model (3D) [GitHub]

Phase 2: Edge Processing Calibration: Phase 2 focuses on calibrating edge devices to effectively filter out noise and non-leak signals, leveraging AI for accurate leak detection.

Phase 3: Adaptive Monitoring and Spatial Analysis: Phase 3 introduces adaptability to the monitoring process, reallocating sensors as needed and employing advanced spatial analysis models for improved leak detection capabilities.

Phase 4: Predictive Analytics and Maintenance Integration: Phase 4 aggregates system data to power predictive analytics, scheduling proactive maintenance and preventing potential leaks before they occur.

References

[1] Blanco, V., & Martínez-Antón, M., Optimal Coverage-Based Placement of Static Leak Detection Devices for Pipeline Water Supply Networks.
[2] Zhao, Q., et al., Estimating Irregular Water Demands with Physics-Informed Machine Learning to Inform Leakage Detection.
[3] Liu, S., et al., Leak Detection in Water Distribution Networks Based on Deep Learning and Kriging Interpolation Method.